Machine Learning–Driven Hospital Resource Optimization and Demand Forecasting

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Abstract*— The optimal utilization of hospital resources is a big challenge for hospitals today. The increasing patient load, lack of infrastructure, and unpredictable admissions are some of the factors contributing to this challenge. This results in overcrowding and inefficient utilization of the hospital's resources. In this context, this paper proposes a machine learning-based hospital resource optimization and patient demand forecasting system. The proposed system helps to optimize the hospital's resources intelligently. The system framework comprises four sub-modules: patient demand forecasting with bed optimization, staff attendance and scheduling analysis, medicine demand forecasting with accident surge analysis, and illness trend forecasting using environmental data. The proposed system uses various machine learning algorithms such as Temporal Fusion Transformers, Random Forest, and XGBoost on the existing hospital and environmental datasets. The experimental results show the effectiveness of the proposed system. The proposed system is a good solution for optimal hospital resource management.

Keywords— *Resource Optimization, Demand Forecasting, Machine Learning, Predictive Analytics, Healthcare Management, Decision Support Systems*

I. Introduction

Healthcare organizations globally are under pressure to deliver quality patient services with scarce resources. With a growing world population, aging citizens, and a rise in chronic and infectious diseases, hospitals are overwhelmed with a surge in patient admissions. Inefficient utilization of scarce resources results in longer waiting times, higher costs of operation, and compromised quality of patient services, especially in resource-scarce systems [15]. A hospital must strike a balance between patient services and scarce resources.

Uncontrollable increases in patient admissions due to seasonal diseases, public health crises, and accident-related admissions cause overcrowding and inefficiencies in hospitals. Emergency departments operate at optimal capacities peak periods, resulting in treatment delays and patient dissatisfaction [3]. These problems underscore the limitations of traditional hospital management practices in dealing with changing healthcare environments.

Traditional management practices employ manual planning and use historical data, which is not flexible enough to respond to changing patient inflow and resource availability in a timely manner. There is a lack of precise predictions and optimizations to facilitate efficient decision-making.

With recent developments in data analytics and machine learning, it has now become possible to predictably model hospital environments. This research aims to develop a framework for a machine learning-based system to facilitate demand forecasting, optimization, and decision-making to improve hospital efficiency. The proposed system aims to reduce waste, improve resource utilization, and minimize patient waiting times for proactive healthcare management.

II. Background and Literature Review

The growing demand in healthcare and limited resources have put pressure on hospital management systems. The traditional approach to hospital management using historical averages and manual planning techniques is not flexible enough to cope with dynamic patient demand. Recent advancements in data analytics and machine learning offer more promising solutions to hospital resource optimization and demand management.

A. Staff Attendance Analysis and Patient-Doctor Schedulimg

Effective doctor scheduling is critical for minimizing waiting times and enhancing service quality. In this context, Paramadevan [7] used linear programming for scheduling physicians. Fernando [10] discussed the inefficiencies in hospital queuing systems. Literature on appointment systems in Sri Lanka indicates the effectiveness of structured scheduling for enhancing patient flow [8], [9]. Nevertheless, there is a lack of real-time visualization tools for scheduling doctor availability.

B. Medicine Demand Forecasting and Accident Surge Detection

Medicine Demand Forecasting and Accident Surge Detection

Research on pharmaceutical supply chains in Sri Lanka focuses on forecasting medicine demand for enhancing medicine availability [11], [13]. The SWASTHA system facilitates enhanced coordination through digital supply management [12]. In addition, accident-related factors have a profound influence on accident rates [14]. Nevertheless, there is a lack of a comprehensive framework for accident-related factors.

C. *Patient Demand Forecasting and Optimization*

Previous research used various statistical and machine learning techniques for forecasting admissions and bed utilization. Jones et al. focused on forecasting patient volume for better resource planning [3]. Advanced techniques such as Temporal Fusion Transformers [1] and Prophet [2] have been used to handle seasonal and temporal patterns. However, most forecasting techniques are not linked with optimization techniques. Bed management is a challenge, and most optimization techniques, including linear programming [4] and [5], have used static constraints.

D. *Illness Trend Forecasting and Outbreak Detection*

It is affected by environmental factors like rainfall and temperature. Though models like SARIMA and LSTM are widely used, the performance is better in Random Forest models. Most of the existing research in Sri Lanka has focused on individual diseases, while multi-disease forecasting systems are not widely implemented.

Table 1: Literature Review

Component	Prior Work Focus	Identified Gap	Proposed Contribution
Staff Attendance & Scheduling Optimization	Scheduling and queuing analysis	No real-time visualization	Interactive availability with decision support
Medicine Demand Forecasting & Accident Surge Detection	Supply management and accident factor analysis	No integrated prediction	Random Forest + XGBoost for adaptive demand planning
Patient Demand Forecasting & Optimization	ML-based demand prediction optimization models	Separate forecasting and optimization	Integrated TFT + LSTM+MILP for dynamic allocation
Illness Trend Forecasting & Outbreak Detection	Disease prediction models (RF)	Single-disease focus	Multi-disease forecasting with environmental data

E. *Research Gaps and Motivation*

The majority of studies focus on individual elements rather than a complete system, making it difficult for real-time adaptability. This research aims to present a unified framework of machine learning, including forecasting, optimization, and predictive analytics, for proactive hospital management.

III. METHODOLOGY

The proposed Machine Learning-Based Hospital Resource Optimization and Demand Forecasting Framework follows a modular data-driven approach to tackle some of the critical issues in hospitals. The proposed system will incorporate predictive analytics, optimization algorithms, and Explainable Artificial Intelligence to assist in proactive decision-making. The proposed methodology comprises four different components, as depicted in Fig. 1.

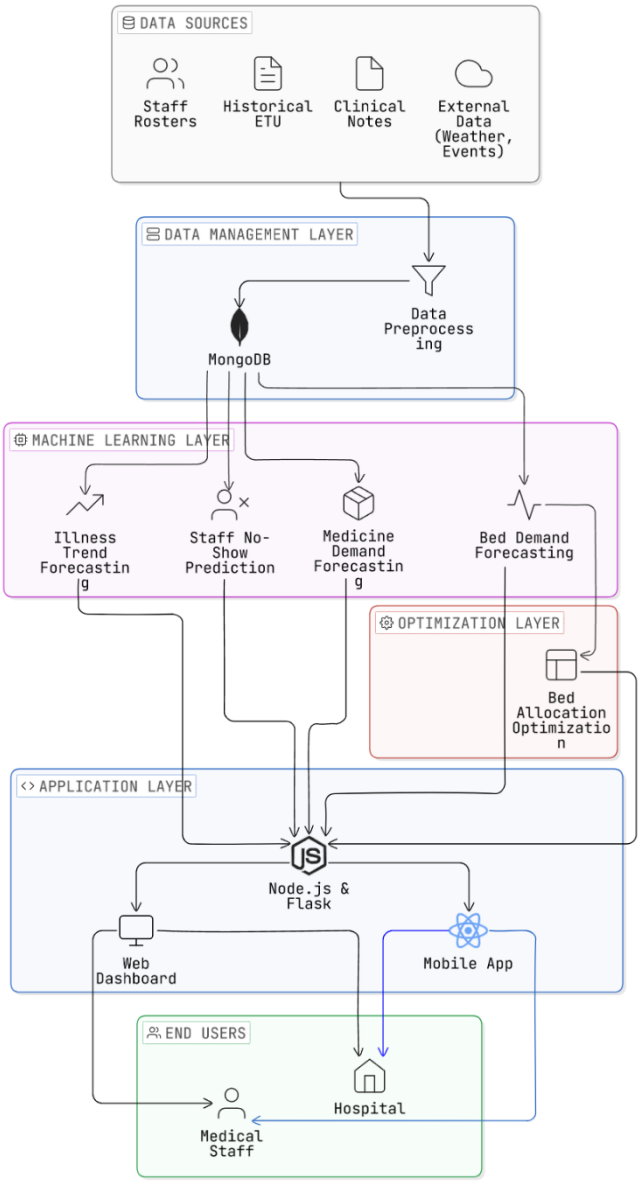

Fig. 1: Methodology Workflow

A. *Staff Attendance Analysis and Patient-Doctor Schedulimg*

This module is designed to minimize the inefficiencies that are created by the non-attendance of patients. Machine learning models are utilized to predict the probabilities of non-attendance. Based on the predictions, the system provides recommendations.

The Explanable AI is also incorporated in the system to provide more information on the factors that are affecting the predictions. The Explanable AI provides more information on the attendance history and the appointment timing. By ensuring

that the attendance is in sync with the doctor's availability, the system optimizes the staff utilization and reduces the idle time in the outpatient department.

B. *Medicine Demand Forecasting and Accident Surge Detection*

This feature enhances medicine inventory planning through predictive analytics techniques. The feature includes a Random Forest model that forecasts regular medicine demand using historical data on medicine usage. Moreover, an XGBoost model is used to analyze accident admission data to identify potential accidents.

In case accident surges are detected, predicted demand for medicines is adjusted to accommodate emergency cases. Otherwise, predicted demand is based on normal usage patterns. This feature improves medicine inventory management in emergency situations while reducing overstocking in normal cases.

C. *Patient Demand Forecasting and Optimization*

This module forecasts short-term ETU patient demand using machine learning time-series models. The forecasted demand is then used with a Mixed Integer Linear Programming (MILP) model to optimize the allocation of beds considering capacity and cost constraints.

Let DDD denote the predicted demand. The decision variables are used to allocate patients to ETU beds, normal beds, surge beds, and external beds. The objective is to minimize costs while preferring normal beds over surge beds or external beds. Constraints are included to ensure that the allocation is within capacity constraints, considering dynamic surge constraints for real-time adaptability.

D. *Illness Trend Forecasting and Outbreak Detection*

This component has a predictive capability for climate-sensitive disease trends based on environmental factors such as rainfall, temperature, and humidity. This is based on a Random Forest algorithm, which can handle non-linear relationships and also includes lag features to mimic reality in disease transmission.

The system can provide both weekly and monthly forecasts to meet short-term and long-term needs. With early warning of outbreaks, proactive allocation of resources such as manpower and medicine preparation can be undertaken by hospitals.

IV. Dataset Description and Experimental Setup

A. *Authors and Affiliations*

The proposed framework is evaluated using four anonymized datasets, corresponding to the four components of the system. All the datasets were first stored within the MongoDB data layer prior to training the models. There was no personally identifiable information (PII) used.

1) *Staff Attendance & Patient Scheduling Dataset*

A structured multi-table data set (Doctors, Weekly_Sessions, Appointments, Exceptions, OPD_Status) has been created to represent the doctor availability and OPD conditions. The data set consists of around 10,000 records for each table, and it has been used for training scheduling and attendance-related models.

2) *Medicine Demand Forecasting & Accident Surge Detection Dataset*

Two structured datasets were used:

- daily medicine usage records (Date, Medicine_Name, Qty_Used)*;*

containing ~16,500 records (2022–2026) from pharmacy and ETU logs. The dataset includes 11 medicines, with four frequently used ETU medicines selected for forecasting.

- accident-related admission records (Date, Accident_Type, No_of_Cases, Injury, Medicines_Used)*;*

containing ~10,000 records across four major accident categories used to train the surge detection model.

3) *Hospital Bed Demand Forecasting and Optimization Dataset*

The dataset includes shift-level ETU records with admissions, discharges, occupancy, capacity, and contextual attributes (DayOfWeek, IsHoliday, SpecialEvent, Weather, PublicTransportStatus, OutbreakAlert, Gender). Approximately 21,918 records from years 2016-2025 were used for training the TFT and LSTM models, as well as the MILP optimization module.

4) *Illness Trend Forecasting and Outbreak Detection Dataset*

The dataset provides disease-level case counts for 11 diseases with environmental factors such as rainfall, temperature, humidity, and lockdown. It comprises 120 records per disease for a period of 10 years, from 2016 to 2025, totaling ~1,320 records, thereby facilitating disease trend prediction using the Random Forest algorithm.

B. *Experimental Prortocol Preprocessing*

Data Preprocessing: This included removing duplicate values, handling missing values, encoding, and normalizing numerical values (for deep learning models). For time-series data, it included arranging data in chronological order and converting it to regular intervals, either monthly or per shift.

- Train–Test Split

Time-series forecasting tasks (ETU admissions, medicine demand, illness trends) used chronological 80/20 splits to prevent data leakage. Classification tasks (no-show prediction) applied record-level stratified splits to preserve class balance.

- Evaluation Metrics
    - Forecasting: MAE, RMSE, MAPE
    - Classification: Accuracy, Precision, Recall, F1-score, Confusion Matrix
    - Optimization: Bed utilization rate, overcrowding incidents, surge-bed activation count, and external transfers

- Baselines

Statistical models like ARIMA, SARIMA, and Random Forest Regressor were employed for baseline forecasting. Deep learning models like LSTM and TFT were compared against these. Models like Random Forest and Gradient Boosting were employed for baseline classification. Optimization techniques employed include comparing MILP results against a rule-based bed allocation strategy.

- *Implementation Environment*

For example, the development of the model employed Python as a programming language for PyTorch, TensorFlow, and Statsmodels. Deployment of the model employed React, React Native, Node.js, Flask, and MongoDB. The training of the model employed Google Colab for GPU acceleration, especially for deep learning models.

Table 2: Summarization of the datasets

Dataset	Granularity	Key Fields	Used For
Staff Attendance / Appointments	Daily + appointment-level	doctor_id,date,is_present, appointment time, outcome	Multi Output Regressor
Medicine usage dataset Accident admissions dataset	Daily Daily	Date, Medicine_Name, Quantity_Used Date, Accident_Type, Number_of_Cases, Injury, Medicines_Used	Random Forest Regressor based medicine demand forecasting XGBoost based accident surge detection
ETU Bed / Demand	shift-level	admissions, discharges, occupied beds, capacity, Gender, DayOfWeek, IsHoliday, Weather, PublicTransportStatus, OutbreakAlert	TFT/LSTM + MILP optimization
Illness trend and outbreak detection	Monthly	Disease, patient count, rainfall, temperature, humidity, 14 day rainfall lag	Random forest regressor

V. RESULTS AND DISCUSSION

A. *Staff Attendance Analysis and Patient–Doctor Scheduling*

The performance of the no-show prediction model was evaluated using Accuracy, Precision, Recall, and F1-score, where the model performed with balanced results for all four metrics. This model can effectively detect no-show appointments with high risk, thereby providing an opportunity for proactive actions, i.e., automatic reminders, rescheduling, and slot allocation, which can improve the efficiency of doctors, reduce idle time, and improve the overall efficiency of the outpatient department.

Table 3 : Model Performance Comparison

Metric	Accuracy	Precision	Recall	F1-score
Value	0.81	0.78	0.74	0.76

The above confusion matrix also proves that there is good classification accuracy because there is a high number of correctly classified show and no-show instances over different time slots and types of patients. There is some classification error in this model, but it is due to the unpredictable nature of patients' behavior patterns rather than errors in classification models. However, this model provides consistent predictive accuracy over different scenarios.

Overall, predictive analytics in scheduling decisions improves the efficiency of operations in healthcare organizations while also allowing more effective and responsive appointment strategies.

B. *Medicine Demand Forecasting and Accident Surge Detection*

- Medicine Demand Forecasting Performance

The performance of the medicine demand forecasting model was evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Weighted Absolute Percentage Error (WAPE).

Table 4 : Performance Comparison for Medicine Demand Forecast

Medicine	MAE	RMSE	WAPE (%)	Accuracy
Inj. Hydrocortizoe 100mg	3.22	4.82	8.95	
Inj. Ompraole	0.87	2.24	3.23	
Inj. Atropine	0.27	1.01	2.12	
Inj. Aderenalin 1:1000	1.17	2.22	8.52	
Overall Model	0.64	1.84	4.55	**95.45%**

From the results, it can be seen that the Random Forest model has been able to capture the trend of medicine consumption for the selected emergency medicines. With precise predictions, hospitals can easily plan for routine medicine demands.

- Accident Surge Detection Performance

The model was evaluated using regression and classification metrics to measure its ability to detect sudden increases in accident-related cases.

Table 5: Performance Comparison for Accident Surges

Metric	MAE	RMSE	Accuracy	Precision	Recall	F1-score
Value	254.30	314.14	0.81	0.82	0.99	0.90

Overall, integrating medicine demand forecasting with accident surge detection improves emergency preparedness and helps healthcare institutions maintain adequate medicine inventories during high-demand periods.

C. *Temporal Hospital Bed Demand Forecasting and Optimization*

- Forecasting Performance Evaluation

Forecasting performance was evaluated using MAE, RMSE, and MAPE. The Temporal Fusion Transformer (TFT) achieved the lowest error values, outperforming LSTM and effectively capturing temporal patterns in admission data. Minor increases in MAPE are due to low actual counts at shift level. Predicted values closely matched actual trends, confirming reliable short-term forecasting performance.

Table 6 : Forecasting Performance

Model	MAE	RMSE	MAPE (%)
LSTM	0.995	1.213	-
TFT	0.838	1.115	37.13

- Optimization Impact Evaluation

The MILP-based allocation improved bed utilization and reduced overcrowding and external transfers compared to rule-based methods, enabling more efficient resource distribution.

Table 7 : Optimization Performance Comparison

Metric	Baseline	Proposed (TFT + MILP)	Improvement
Average Bed Utilization	82%	91%	+9%
Overcrowding Incidents (per month)	14	6	–57%
Surge Bed Activation Rate	18%	12%	Reduced and Strategically Controlled
External Transfers	9	3	–66%

Overall, integrating forecasting with optimization enhances decision-making, reduces congestion, and supports proactive hospital resource management.

D. *Illness Trend Forecasting and Outbreak Detection for Hospitals*

Comparative experiments with SARIMA, Prophet, LSTM, and Random Forest models were carried out using MAE, RMSE, and MAPE metrics to measure the performance of the models. The results show that the Random Forest model outperforms both statistical models and deep learning models by successfully modeling the non-linear relationship between environmental factors and diseases, especially rainfall lag effects. The LSTM model performed satisfactorily, though it requires large data to make predictions. Prophet performed satisfactorily, though it failed to make precise predictions for short-term forecasting..

Table 8 : Performance Comparison for Illness Trend Forecasting

Model	MAE	RMSE	MAPE %
SARIMA	12.45	28.4	18.9
LSTM	8.54	10.20	12.3
Prophet	10.12	12.85	14.7
Random forest (Weekly)	0.82	1.25	6.4
Random forest (Monthly)	3.15	4.10	9.2

The Random Forest model performed consistently with both weekly and monthly forecasting, making it an appropriate model for outbreak forecasting. This is due to its capability to make precise predictions, which is essential for proactive planning, especially for climate-sensitive diseases like Dengue and Leptospirosis. This is particularly important for strategic planning, especially for diseases like Dengue and Leptospirosis.

E. *Discussion of Findings*

The results demonstrate the power of deep learning and ensemble techniques in modeling complex healthcare patterns. TFT has shown superior performance in patient demand forecasting, while RF and XGB performed better in medicine demand prediction and accident surge detection.

The integrated framework has demonstrated the power of integrating forecasting, optimization, and predictive analytics in hospital operations. Improved accuracy in forecasting helps in better bed allocation, reduced congestion, and enhanced emergency preparedness.

In conclusion, the system acts as a comprehensive framework for hospital decisions, helping to transition from reactive to proactive hospital management.

VI. CONCLUSION AND FUTURE WORKS

This research presents a framework for optimizing hospital resources and forecasting demands using machine learning techniques to solve some of the most pressing challenges facing hospitals. This framework combines the benefits of predictive analytics, mathematical optimization, and data-driven decision-making for the proactive management of hospital resources, staff, medicine, and diseases. The framework's modularity allows for flexibility and expandability, making it suitable for implementation in a real-world hospital setting, especially in a resource-scarce environment.

The experimental results show a clear improvement in forecasting, resource management, and decision-making compared to traditional reactive methods. The integration of various tools within a single platform allows for a more efficient hospital operation.

Although this research presents several benefits, there are some limitations. It relies on historical data and uses simulation for evaluation. Future research will be conducted on real-time integration, multi-hospital coordination, reinforcement learning techniques, and notification systems

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